

Root Grip Erosion Rescue

Can roots hold a slope together in a storm?

At a glance: Can roots hold a slope together during a storm? Plan about 60 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

BY THE END OF THIS ACTIVITY, EACH STUDENT CAN:

- Explain the core science of this challenge: roots, erosion, soil stability, slope design (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- Trays: 2 per team (one bare slope, one rooted slope), 8 plus 2 spare
- Soil: about 4 to 6 cups packed per team across the two trays (bring 2 to 3 gal for the camp; source locally, not a stocked buy item)
- String or yarn roots: cut into 4 to 6 in lengths, about 12 to 20 per team (2 to 3 balls; source locally)
- Craft sticks: 4 to 6 per team to prop and anchor the slope (1 pack, reused; source locally)
- Watering cups: 1 marked pour cup per team (4 plus 4 spare; source locally, missing from the TTB-01 buy line)
- Nitrile gloves: 1 pair per student (shared PPE box, already bought)
- Water and a slop bucket: tap water, about 1 gal of pours per team; pour muddy runoff into the bucket, not the drain

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

2 Mark each watering cup with a fill line (about 1 cup) and set a fixed pour height, stage the soil, string/yarn roots, craft sticks, a fill jug, and a slop bucket, and put out the scoring sheet before testing begins.

3 Put out nitrile gloves and a containment tray at each station before any soil is handled. No goggles are needed: this is wet soil and a poured cup of water, not a chemical or splash hazard. Sieve and lightly dampen the soil to keep dust down.

01 Load the activity slides and ready the projected timer.

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Can roots hold a slope together during a storm?" Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:38

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams pour the standard storm (the marked cup from the set height) on the bare slope and the rooted slope over the tray, judge or weigh the soil retained on each, and log the baseline before changing anything.

Phase 6 Redesign

Each team changes exactly one variable in the root network (more roots, deeper roots, or a denser pattern), predicts the effect, and re-pours the standard storm to compare against baseline.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, then reset the station: pour muddy runoff into the slop bucket (not the sink), return soil to the soil bin, rinse and dry the string/yarn roots and craft sticks, rinse and wipe the trays and cups, dry the table and floor, then remove gloves and wash hands.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: Both slopes lose about the same amount of soil on the first pour. **Fix:** Check that both slopes are the same angle and packed the same, that the roots are pushed deep into the rooted slope (not just laid on top), and that the same marked cup is poured from the same height; then log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to roots anchor soil.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why did the rooted slope hold more soil?
- How would deeper roots change the result?
- Where is this used to protect real hillsides?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Soil retained	40
Runoff control	25
Design explanation	20
Cleanup	15
Total	100

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES EPA green infrastructure and erosion lessons